Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/31/2007
PI Name:	Minna, John D. M.D.		
Project Title:	NSCOR: Lung Cancer Pathogenesis and HZE Particle Exposure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-8593	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 NSCOR Space Radiation NNH04ZUU002N
Start Date:	03/31/2005	**End Date:**	03/31/2010
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Start/end dates changed to 3/31/2005-3/31/2010 per grant documents from PI (12/06)		
Key Personnel Changes/Previous PI:	Dr. Story has become the Lead of Project #1 with Dr. Gazdar becoming the Co-Lead. This is because the majority of the findings appear to revolve about gene expression and genetic changes after HZE irradiation, and because of the large amount of HBEC cell survival studies that are needed that Dr. Story is leading. Dr. Minna and Dr. Shay have assumed responsibility for the Cell Culture Core. This occurred because of the focus and need for expanding human bronchial epithelial cells for nearly all NSCOR related experiments and their own labs expertise in this. Dr. Story has assumed responsibility for all of the Expression profiling and proteomics work of the genomics core because of other commitments of Dr. Hinging Zhao who had been a Core Co-lead for proteomics. Dr. Zhao will still be available for proteomics consultation as initially planned. Andres Roig, M.D. and Oliver Delgado (Graduate student) have joined Projects #3 and #4 in the “TBD” positions. They are both fully trained, certified and have worked at BNL for the NSRL7 run.		

COI Name (Institution):	Abrams, John (The University of Texas Southwestern Medical Center) Burma, Sandeep (The University of Texas Southwestern Medical Center) Caetano, Raul (The University of Texas School of Public Health) Chen, David (The University of Texas Southwestern Medical Center) Gazdar, Adi (The University of Texas Southwestern Medical Center) Girard, Luc (The University of Texas Southwestern Medical Center) Harris, T. Robert (The University of Texas School of Public Health) Richardson, James (The University of Texas Southwestern Medical Center) Shay, Jerry (The University of Texas Southwestern Medical Center) Story, Michael (The University of Texas Southwestern Medical Center) Wright, Woody (The University of Texas Southwestern Medical Center) Minna, John (The University of Texas Southwestern Medical Center)
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Task Description:	The University of Texas Southwestern Medical Center (UTSW) NSCOR proposal focuses on the development of quantitative dose risk estimates following HZE particle irradiation for the development of key genetic, epigenetic, gene expression, and cellular functional changes in the multistep pathogenesis of lung cancer in both new human bronchial epithelial cell (HBEC) and transgenic mouse models of lung cancer. The risk of developing these changes will also be compared in these models to that of X-irradiation. These dose risk assessments will measure events of both radiation-induced promotion (modification of proliferation kinetics of already-initiated cells) as well as radiation-induced initiation (mutational) events. 3D organotypic culture and animal models will also allow measurement and risk estimation of “bystander” effects. These individual risk estimates can then be combined into a model (such as a two-stage clonal expansion model) for overall risk of developing lung cancer from exposure to galactic cosmic radiation. To achieve these goals, UTSW has assembled a team of scientists who are leaders in the study and translation application of the molecular pathogenesis of lung cancer (Minna, Gazdar, Shay); in radiobiology and DNA repair (Chen, Story); a panel of expert Internal and External Advisors (from other NSCORS and lung cancer research experts); as well as expert consultants and collaborators. This team has developed a novel immortalized HBEC system that can be genetically manipulated, studied in 2D monolayer and 3D organotypic cultures. Using these HBECs we have preliminary studies with HZE particle irradiation showing that further progression toward lung cancer can be detected and quantified, that specific expression profiles for HZE irradiation exist, and that HBECs genetically manipulated to progress part of the way toward malignancy are even more sensitive to HZE-induced functional changes. In addition, we have developed plans for novel transgenic mouse models to monitor HZE particle induction of lung cancer in the whole animal in real time. The NSCOR has 4 Projects: 1. Genetic and epigenetic changes in human bronchial epithelial cells following exposure to HZE particle irradiation.; 2. Effect of HZE particles on DNA damage-sensing and repair pathways in human lung epithelial and fibroblast cells; 3. Effect of HZE particle irradiation on functional progression of human lung cancer at the cellular and organotypic level; and 4. Effects of HZE particles on the development of lung cancer in vivo in novel mouse models. These projects are supported by 4 Cores: Administrative; Cell Culture; Expression Profiling and Proteomics; and Biostatistics and Bioinformatics. In addition, with funding from the Department of Energy (DOE) contributing to this NSCOR, studies of low-LET and low dose rate radiation in the same model systems are also being undertaken. The HZE particle irradiation will be done at Brookhaven National Lab, and the biostatistical analysis shows the experiments are powered to help achieve NASA mandated risk estimate confidence levels. Finally, since this group has substantial UTSW institutional commitments, and holds a NCI Special Program of Research Excellence (SPORE) in Lung Cancer and a Center for the NCI Early Detection Research Network (EDRN) program, there will be great synergism and additional resources available for the successful completion of this proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Lung cancer is the leading cause of cancer death in both women and men in the USA and in the Western world. While lung cancer is caused by smoking exposure in 85% of people, because of the large numbers the remaining 15% of cases arising in life time never smokers (~25,000 patients per year) is a sizable health problem in the USA. Likewise, 50% of all new lung cancer cases occur in former smokers (quitting over 5 years before). In all of these scenarios a major underlying question has been the role of other environmental carcinogens especially environmental radiation. One source of this is radon gas in the environment including radon gas exposure in homes with alpha particles (which represent high LET radiation). Thus, the interaction of radiation in the environment on Earth including from alpha particles in the genesis of lung cancer and quantitation of the effect of such radiation on lung epithelial cells is of major health importance. Radiation is a known carcinogenic influence but the molecular events in lung epithelial cells in response to radiation need to be determined. A key element of this is the impact of radiation on lung epithelial cells with preneoplastic lesions. Our NSCOR uses innovative technology of human bronchial epithelial cells (HBECs) developed by us (coming from 36 individuals) which have also been engineered to have known oncogenic preneoplastic changes (such as oncogenic KRAS and knockdown of p53). Our studies are providing quantitative data following high LET irradiation on these HBECs. The primary benefits from this work will be the development of quantitative risk assessment models for astronauts of high LET radiation in space. However, the results will also have major impact and benefit to life on Earth includes new knowledge of the effects on human lung epithelial cells of irradiation in terms of quantitative genetic and epigenetic changes and gene expression changes following irradiation. As part of this will be the quantitative response of DNA repair pathways to radiation including specific DNA repair components such as DNA repair enzymes and signaling pathways in lung epithelial cells. We are also determining quantitative changes in biologic phenotypes of lung epithelial cells in response to radiation which include colony formation in liquid and soft agar (anchorage independent growth) and differentiation and invasion in 3 dimensional cultures, and finally growth in vivo as either xenografts (for human cells) or endogenous lung cancers (for mouse models of lung cancer). From all of the above information we will identify novel molecular biomarkers (mRNA, DNA, protein) for lung carcinogenesis. Finally, we are learning about inter individual variation of the response of HBECs to radiation with respect to the various genetic, epigenetic, mRNA and protein expression, and signaling pathway changes.

Task Progress:

Project 1. Genetic and epigenetic changes in human bronchial epithelial cells following exposure to HZE particle irradiation. Michael Story, Ph.D. Lead, Adi F. Gazdar, M.D. Co-Lead, John D. Minna, M.D. Co-Lead
The overall goals of project 1 are to establish both the low LET and the HZE radio-response of human bronchial epithelial cells (HBECs) that have been genetically manipulated, or not, with a variety of oncogenic changes that are often seen during the oncogenic progression towards lung carcinogenesis. These changes include: the downregulation of p53; the upregulation of oncogenic KRASV12; and the overexpression of wild type and mutant EGFR, as examples. These cell lines represent in vitro equivalents of preneoplastic lesions and even with multiple changes, these manipulated HBECs show only part of the malignant phenotype and are not capable of forming tumors in immunodeprived mice. Characterization of these cell lines includes standard survival analysis following radiation exposure, gene expression analysis, miRNA analysis, protein expression, DNA methylation, clonogenicity in soft agar, and tumorgenicity. Our intent is to follow irradiated cell cultures up to 4 months post-irradiation in order to examine these endpoints in cells that had survived a series of low LET exposures or specific HZE particle exposures in order to determine the effect the radiation exposure has on the oncogenic process. We will also examine inter-individual responses in those endpoints found above that enhance the oncogenic process. Our initial experiments have focused on clonogenic survival after low and high LET radiation exposures. As expected, we found that Fe56 particles are, per unit dose, far more effective at cell killing than gamma-ray exposures. Furthermore, the differences in radiosensitivity seen with low LET exposures between normal cells and cells with mutant EGFR are eliminated after Fe exposure. Si particle exposure, even though at the same energy as the Fe particles produced an intermediate killing effect. This result corresponds well with the ionization intensity differences seen via foci formation in HZE irradiated cells as determined by Project 2. Experiments to determine transformation frequency as a function of radiation exposure type are ongoing (see Project 3). Thus far, no difference has been seen four months after a 1Gy Fe exposure. This dose is highly effective at cell killing and may have eliminated cells that at lower doses would have transformed. Therefore, we are reducing the dose accordingly and re-examining transformation potential. We will continue to fill in the gaps in data for low LET exposure, and now that we have Fe and Si exposed samples, we have started the analysis of the molecular and biochemical endpoints described above, in particular, gene expression analysis. The initial data on gene expression suggest that cells do respond differently to Fe particle exposure as compared to low LET exposures. This response is likely not dose dependent but may reflect an altered kinetic response to the radiation exposure.

Project 2. Effect of HZE particles on DNA damage-sensing and repair pathways in human lung epithelial and fibroblast cells. David Chen, Ph.D. Project Lead, Sandeep Burma, Ph.D. Co-Lead

With the ultimate objective of estimating cancer risks to humans from HZE particles, in this project we investigate the DNA damage responses of human cells to HZE, the link between these responses and cancer predisposition being very well established. We will take advantage of recent advances in the study of molecular mechanisms of DNA damage sensing, damage signaling, and DNA repair following low LET radiation to verify the hypothesis that HZE particles generate a different and/or unique radiation response in mammalian cells as compared to X-rays. As we are specifically interested in risks related to lung cancer development upon exposure to space radiation, we use immortalized normal human bronchial epithelial cell lines (HBECs) and three types of lung fibroblasts (WI38, IMR90 and MRC5) as model systems. Although these HBEC cultures (2D cultures) will be used throughout the project, we will also use normal or premalignant monolayer cultures (2D) and organotypic cultures (3D cultures) to examine these responses in a more in vivo tissue-like setting. The proposed research should reveal fundamental differences in cellular responses between low LET X-rays and HZE exposures in humans and in turn, provide biological markers correlated to lung carcinogenesis for the development of risk assessment and radiation protection for astronauts during future space missions. The objectives of this project are to examine the complexity of DNA damage induced by HZE relative to gamma rays in 2D and 3D cultures of HBECs and to understand the pathways responsible for repair of the complex damage inflicted by these particles and the consequences of inadequate repair responses. Using 1 GeV Fe and X-rays, we have worked out the early damage-responses, damage-repair kinetics, and persistence of DNA damage in primary human skin fibroblasts (HSFs) in great detail. Both X-rays and 1 GeV Fe particles elicit similar responses at the sites of DNA breaks. However, Fe-induced damage is repaired more slowly and a large fraction of breaks are not rejoined, thereby resulting in the higher relative biologic effect (RBE) of HZE particles. Using charged particles of increasing molecular weights (O vs. Si vs. Fe) we find that the incidence of unrejoined breaks increases with increasing molecular weight presumably due to increasing complexity of the damage induced. Detailed time courses have already been carried out with a panel of HBECs irradiated with 1 GeV Fe particles. The resulting data are currently undergoing analyses. Future plans include characterization of these responses in lung fibroblasts and epithelial cells. We have already started analyses of 2D cultures of HBECs that are in different pre-malignant stages of cancer development and in organotypic 3D cultures of lung epithelial cells and fibroblasts.

Project 3. Effect of HZE Particle Irradiation on Functional Progression of Human Lung Cancer. John D. Minna, M.D. Project Lead, Jerry Shay, Ph.D. Co-Lead

This project studies the effect of 2D (tissue culture) and 3D organotypic culture on the response to radiation from high and low LET irradiation. It is important to study the effect of HZE radiation on both the epithelial and stromal (fibroblast) components. Studying the effects of HZE radiation on lung epithelial cells should help elucidate direct mechanisms promoting initial neoplastic transformation as well as progression. Studying the effects of HZE radiation on stromal tissues may uncover other factors contributing to cancer formation and progression such as the disruption of important aspects of cellular cross- talk that promote tissue homeostasis. As described in Project 1, HBECs including HBECs engineered to contain various oncogenic changes were irradiated and then Project 3 tested them for biologic behavior relevant to cancer beginning with studies of anchorage independent growth in soft agar (a characteristic of cancer cells). The HBECs before any changes do not grow in soft agar, but do to varying degrees after these changes. We measured soft agar colony formation 1,2,3, and 4 months after HZE particle irradiation compared to control un irradiated cells. At one month there was increased colony formation in HBECs modified to contain HPV oncogenic E6 and E7 proteins, but not in any of the other 8 HBECs with various oncogenic changes or at other time points. We have also tested the effect of radiation on cells in a 3D (organotypic) system: the fibroblast collagen plug. The lung fibroblast plug closely resembles the stromal pulmonary environment. We irradiated lung fibroblast cells in regular tissue culture and in a collagen plug (3D) arrangement with low and high LET (1 Gy X-ray or 1 Gy HZE Fe56) and studied the DNA damage response and kinetics of repair. After irradiation, the plugs and cellular monolayers were fixed at specific time points (time 0 (control), 2, 12, 24, and 48 hours, and 12 days) and stained using immunofluorescence microscopy for gamma-H2AX as a marker for DNA damage which appears as large discrete foci. Radiation of low and high LET led to damage foci in 90% of cells and the amount of DNA damage was comparable up to 24 hours. However, strikingly, while the low LET radiation DNA damage was resolved by 48 hours, the high LET DNA damage persisted for up to 12 days. We are now testing the 3D cultures with epithelial cells and fibroblasts combined, and we are studying the effects

	of age, gender, dose rate as well as the gene expression profiles. Project 4. Effects of HZE Particles on the Development of Lung Cancer in Novel Mouse Models. Jerry Shay, Ph.D. Project Lead, James Richardson, DVM, Ph.D. Transgenic mouse models of human lung cancer have the advantage of a similar tumor development pathway compared to humans but with a shorter developmental period. The LA1 K-ras mouse model we have used contains a latent oncogenic K-ras allele integrated in the endogenous K-ras locus. The activation of the oncogenic allele occurs spontaneously preferentially in the lungs as the animals mature. Every animal containing the latent allele develops alveolar type II lineage lung cancer by the age of four to nine months and all animals die at approximately one year from tumor burden. Our primary objective is to determine if these animals, after low and high LET exposure, initiate or progress to cancer at increased frequency. Our long-term goal is to determine if there is an increase, decrease or no change of onset of death after HZE irradiation and ultimately to use this information for modeling risks in humans. Other end points include other cancer types, DNA damage and repair (gamma-H2AX, DNA-PKc) to correlate with the 2D and 3D cell culture experiments in projects 2 and 3. In addition, we will obtain histology and conduct expression profiles of low and high LET irradiated mouse lungs to compare to the cell culture experiments in Projects 1-3. We have conducted low LET (X-ray) at UT Southwestern Medical Center and high LET (Fe56) at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL). 94 animals were irradiated with 0.2 Gy x 5 (1GeV Fe56), 97 animals were irradiated with 1.0 Gy (1GeV Fe56), 110 animals irradiated with 0.2 Gy x 5 (X-ray), and 100 animals irradiated with 1Gy (X-ray). In addition, we have 98 non-irradiated control animals. To examine acute responses (3 hr post fractionated and single dose low and high LET radiation), lung tissue was removed and RNA isolated for gene expression arrays. We are bringing on line a new mouse model of lung cancer using a transgenic mouse with a Tet-inducible mutant EGFR gene that results in lung cancer. Overall these experiments allow us to study the effects of HZE and low LET radiation on the development of lung cancer in a whole animal model.
Bibliography Type:	Description: (Last Updated: 11/20/2025)
Articles in Peer-reviewed Journals	Sunaga N, Gazdar AF, Shay JW, Minna JD. "Multiple oncogenic changes (K-rasV12, p53 knockdown, mutant EGFRs, p16 bypass, telomerase) are not sufficient to confer a full malignant phenotype on human bronchial epithelial cells. " Cancer Res. 2006 Feb 15;66(4):2116-28. PMID: 16489012 , Feb-2006
Articles in Peer-reviewed Journals	Vaughan MB, Ramirez RD, Wright WE, Minna JD, Shay JW. "A three-dimensional model of differentiation of immortalized human bronchial epithelial cells. " Differentiation. 2006 Apr;74(4):141-8. PMID: 16683984 , Apr-2006